

Case Study 2: Forecasting with AR and ARIMA models

Question #1

Ar(1) model on entire historical data, first differencing + ACF

```
> ar1.full <- Arima(revenue.ts, order = c(1,0,0))
> summary(ar1.full)
Series: revenue.ts
ARIMA(1,0,0) with non-zero mean

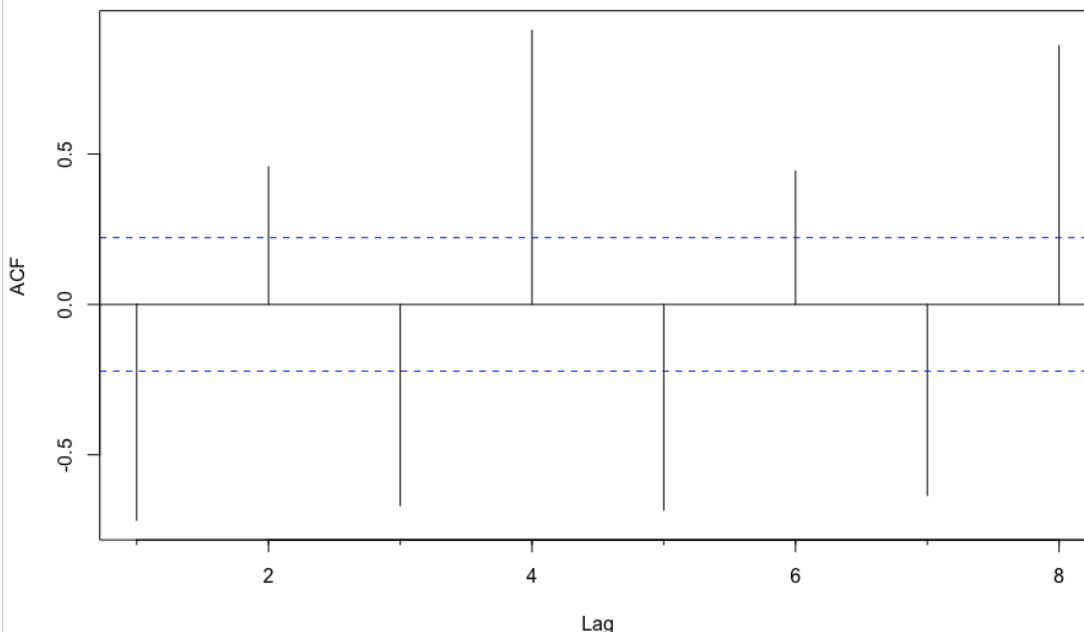
Coefficients:
      ar1      mean
    0.9485 127211.70
s.e.  0.0399  17576.25

sigma^2 = 96984434: log likelihood = -838.64
AIC=1683.28  AICc=1683.6  BIC=1690.39

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 966.264 9722.609 8265.906 0.2226992 6.735426 1.69258 -0.6625217
```

The AR(1) model fitted to the entire historical Walmart revenue series produces an autoregressive coefficient of **0.9485** with a standard error of **0.0399**. Since the coefficient is large and statistically significant, it indicates strong persistence in the time series. This means that the revenue in one quarter is highly dependent on the revenue of the previous quarter. In other words, Walmart's revenue exhibits strong temporal dependence and is therefore predictable from past values. The coefficient being close to 1 suggests that shocks to revenue persist over time, which is common for large retail revenue series.

ACF of First-Differenced Walmart Revenue



The ACF plot of the first-differenced Walmart revenue series shows several significant spikes outside the confidence bands, particularly at **lag 1 and lag 4**. The significant spike at **lag 1** indicates short-term autocorrelation in quarterly changes in revenue.

The spike at **lag 4** suggests the presence of **quarterly seasonality**, which is consistent with retail sales patterns where the fourth quarter (holiday season) typically shows higher revenues.

Question #2

A regression model with quadratic trend and seasonal indicators was first fitted to capture the long-term growth and seasonal structure of the revenue series. Residual diagnostics showed significant autocorrelation at lag 1, indicating remaining time dependence. An AR(1) model was therefore fitted to the regression residuals. Forecasts from the AR(1) model were added to the regression forecasts to produce a two-level combined forecast. This approach improved forecast accuracy compared to the regression model alone, as evidenced by lower RMSE, MAE, MAPE, and Theil's U.

Previous time series function from case 2 - regression with quadratic trend + seasonality on training set:

```
> reg.quad <- tslm(train.ts ~ trend + I(trend^2) + season)
> summary(reg.quad)
```

Call:

```
tslm(formula = train.ts ~ trend + I(trend^2) + season)
```

Residuals:

Min	1Q	Median	3Q	Max
-4914.3	-1858.3	-600.4	1937.2	7673.1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	81899.443	1338.764	61.175	< 2e-16 ***
trend	1161.865	89.516	12.979	< 2e-16 ***
I(trend^2)	-6.030	1.422	-4.240	8.8e-05 ***
season2	4243.516	1078.843	3.933	0.000241 ***
season3	1750.491	1079.520	1.622	0.110726
season4	14503.194	1080.643	13.421	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2954 on 54 degrees of freedom

Multiple R-squared: 0.9672, Adjusted R-squared: 0.9641

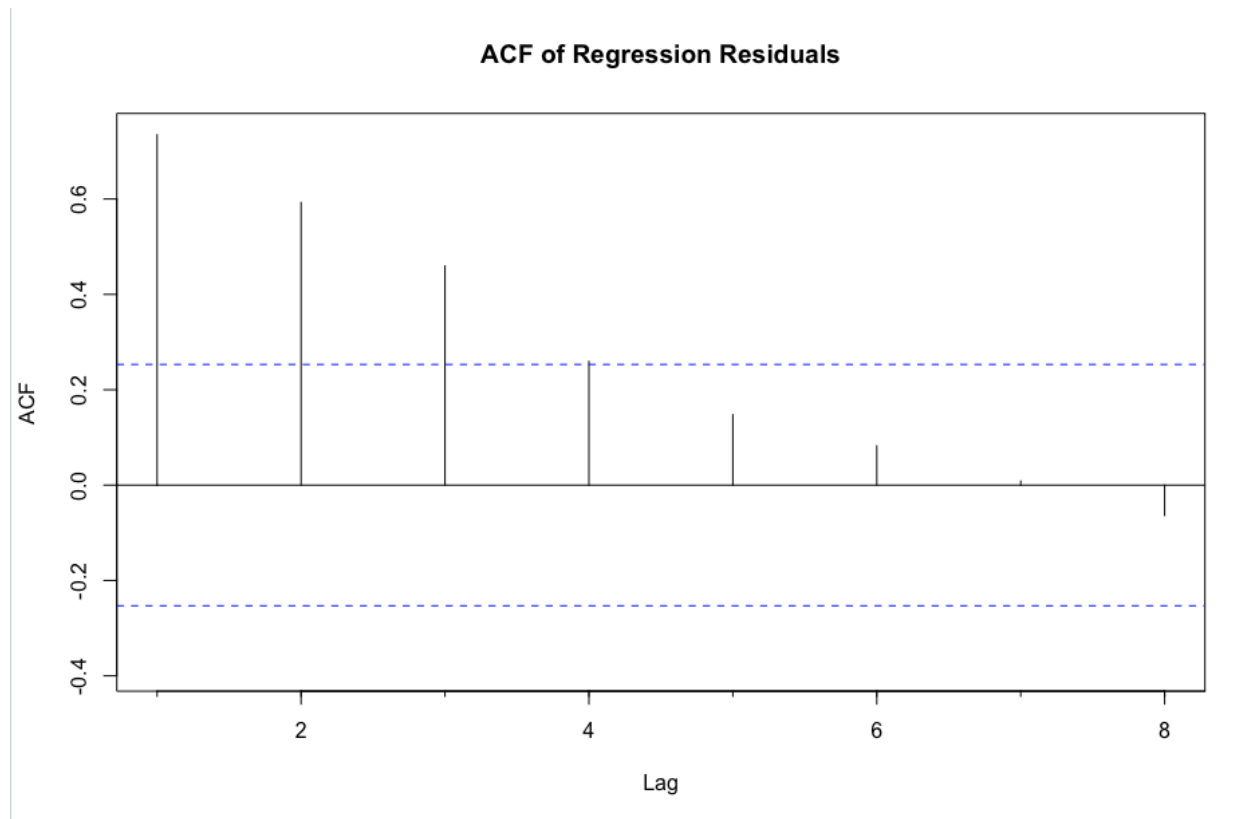
F-statistic: 318.3 on 5 and 54 DF, p-value: < 2.2e-16

Prediction using valid partition:

```
> reg.pred <- forecast(reg.quad, h = nValid, level = 0)
> reg.pred
```

	Point	Forecast	Lo 0	Hi 0
2021 Q1		130334.9	130334.9	130334.9
2021 Q2		134998.6	134998.6	134998.6
2021 Q3		132913.7	132913.7	132913.7
2021 Q4		146062.4	146062.4	146062.4
2022 Q1		131943.2	131943.2	131943.2
2022 Q2		136558.6	136558.6	136558.6
2022 Q3		134425.4	134425.4	134425.4
2022 Q4		147525.9	147525.9	147525.9
2023 Q1		133358.4	133358.4	133358.4
2023 Q2		137925.6	137925.6	137925.6
2023 Q3		135744.2	135744.2	135744.2
2023 Q4		148796.5	148796.5	148796.5
2024 Q1		134580.8	134580.8	134580.8
2024 Q2		139099.7	139099.7	139099.7
2024 Q3		136870.1	136870.1	136870.1
2024 Q4		149874.1	149874.1	149874.1
2025 Q1		135610.1	135610.1	135610.1
2025 Q2		140080.8	140080.8	140080.8
2025 Q3		137802.9	137802.9	137802.9

Residuals of regression model ACF:



Ar(1) model for regression residuals:

```
> ar1.res <- Arima(reg.res, order = c(1,0,0))  
> summary(ar1.res)
```

Series: reg.res

ARIMA(1,0,0) with non-zero mean

Coefficients:

	ar1	mean
	0.8429	321.0298
s.e.	0.0789	1292.2419

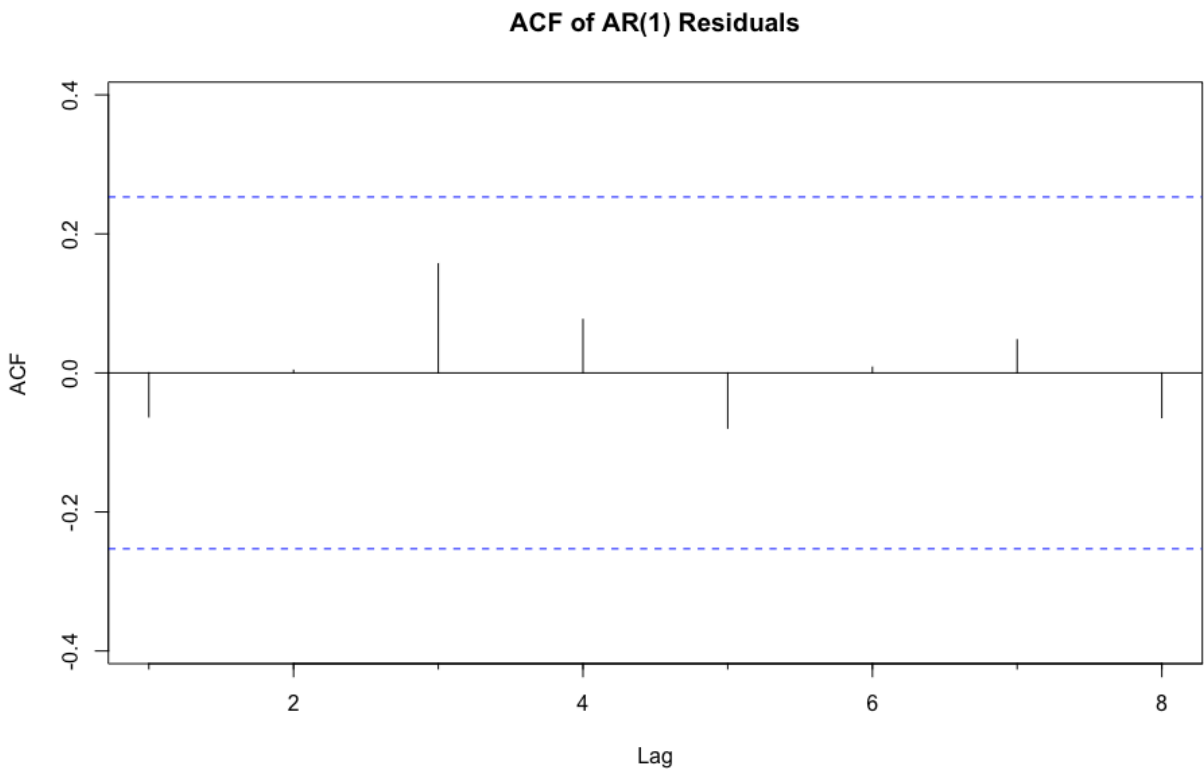
$\sigma^2 = 2970217$: log likelihood = -531.86

AIC=1069.73 AICc=1070.16 BIC=1076.01

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	81.31996	1694.465	1151.531	42.21883	82.14699	0.4754662	-0.06318279

Residuals of residual model ACF:



Beginning of two level forecast for validation partition:

```
> ar1.res.pred <- forecast(ar1.res, h = nValid, level = 0)
> ar1.res.pred
```

	Point Forecast	Lo 0	Hi 0
2021 Q1	6517.8402	6517.8402	6517.8402
2021 Q2	5544.1066	5544.1066	5544.1066
2021 Q3	4723.3803	4723.3803	4723.3803
2021 Q4	4031.6185	4031.6185	4031.6185
2022 Q1	3448.5565	3448.5565	3448.5565
2022 Q2	2957.1137	2957.1137	2957.1137
2022 Q3	2542.8936	2542.8936	2542.8936
2022 Q4	2193.7618	2193.7618	2193.7618
2023 Q1	1899.4908	1899.4908	1899.4908
2023 Q2	1651.4599	1651.4599	1651.4599
2023 Q3	1442.4032	1442.4032	1442.4032
2023 Q4	1266.1966	1266.1966	1266.1966
2024 Q1	1117.6781	1117.6781	1117.6781
2024 Q2	992.4971	992.4971	992.4971
2024 Q3	886.9863	886.9863	886.9863
2024 Q4	798.0549	798.0549	798.0549
2025 Q1	723.0977	723.0977	723.0977
2025 Q2	659.9189	659.9189	659.9189
2025 Q3	606.6677	606.6677	606.6677

Forecast comparison. accuracy of regular vs two level:

```
> valid.df
  Validation_Data Regression_Forecast AR1_Residual_Forecast Combined_Forecast
1      138310      130334.9      6517.840      136852.8
2      141048      134998.6      5544.107      140542.7
3      140525      132913.6      4723.380      137637.0
4      152871      146062.4      4031.619      150094.0
5      141569      131943.2      3448.557      135391.7
6      152859      136558.6      2957.114      139515.7
7      152813      134425.4      2542.894      136968.3
8      164048      147525.9      2193.762      149719.7
9      152301      133358.4      1899.491      135257.9
10     161632      137925.6      1651.460      139577.1
11     160804      135744.2      1442.403      137186.6
12     173388      148796.5      1266.197      150062.7
13     161508      134580.8      1117.678      135698.4
14     169335      139099.7      992.497      140092.2
15     169588      136870.1      886.986      137757.0
16     180554      149874.1      798.055      150672.1
17     165609      135610.1      723.098      136333.2
18     177402      140080.8      659.919      140740.7
19     179496      137802.9      606.668      138409.6

> round(accuracy(reg.pred$mean, valid.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 21639.69 24071.78 21639.69 13.135 13.135 0.814      2.501

> round(accuracy(combined.pred, valid.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 19323.71 22810.83 19323.71 11.592 11.592 0.819      2.362
```

Beginning of two level forecast for entire dataset:

```
> summary(reg.quad.full)
```

Call:

```
tslm(formula = revenue.ts ~ trend + I(trend^2) + season)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-9115.8 -4109.7   299.7  3893.7  9914.2
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 88385.286   1931.622  45.757 < 2e-16 ***
trend        398.904     98.478   4.051 0.000126 ***
I(trend^2)    7.581      1.193   6.355 1.60e-08 ***
season2      4938.372   1557.392   3.171 0.002222 **
season3      2698.682   1557.961   1.732 0.087463 .
season4     14362.138   1579.012   9.096 1.23e-13 ***
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 4924 on 73 degrees of freedom

Multiple R-squared: 0.9619, Adjusted R-squared: 0.9593

F-statistic: 369 on 5 and 73 DF, p-value: < 2.2e-16

Forecasting 9 quarter ahead:

```
> reg.full.pred <- forecast(reg.quad.full, h = 9, level = 0)
> reg.full.pred
```

	Point Forecast	Lo 0	Hi 0
2025 Q4	183176.1	183176.1	183176.1
2026 Q1	170433.4	170433.4	170433.4
2026 Q2	177006.3	177006.3	177006.3
2026 Q3	176416.4	176416.4	176416.4
2026 Q4	189744.7	189744.7	189744.7
2027 Q1	177062.6	177062.6	177062.6
2027 Q2	183696.2	183696.2	183696.2
2027 Q3	183166.8	183166.8	183166.8
2027 Q4	196555.8	196555.8	196555.8

Ar(1) on full data residuals:

```
> ar1.full.res <- Arima(reg.full.res, order = c(1,0,0))
> summary(ar1.full.res)
```

Series: reg.full.res
ARIMA(1,0,0) with non-zero mean

Coefficients:

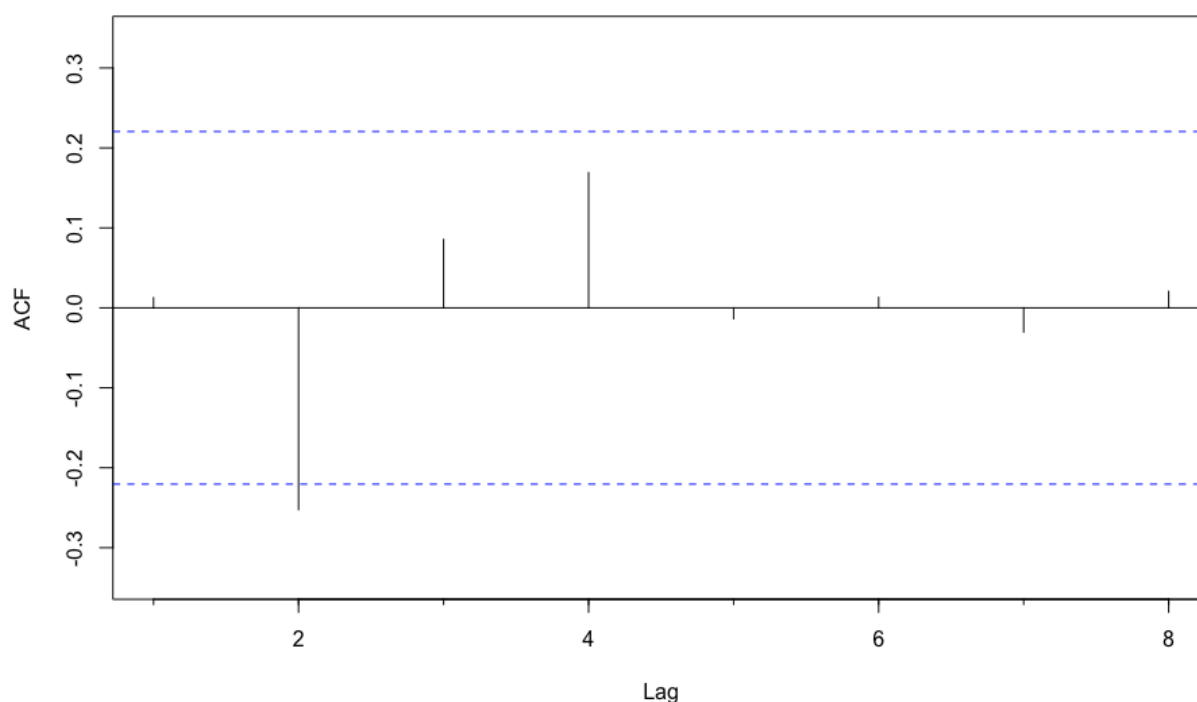
	ar1	mean
	0.9384	91.9726
s.e.	0.0430	3062.3325

sigma^2 = 3998174: log likelihood = -712.6
AIC=1431.2 AICc=1431.52 BIC=1438.31

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	183.3889	1974.071	1502.192	-2070.843	2200.773	0.5240204	0.01266782

ACF of AR(1) Residuals - Full Data



Forecast 9 quarter ahead:

```
> ar1.full.pred <- forecast(ar1.full.res, h = 9, level = 0)
> ar1.full.pred
```

	Point Forecast	Lo 0	Hi 0
2025 Q4	9002.310	9002.310	9002.310
2026 Q1	8453.146	8453.146	8453.146
2026 Q2	7937.828	7937.828	7937.828
2026 Q3	7454.271	7454.271	7454.271
2026 Q4	7000.516	7000.516	7000.516
2027 Q1	6574.727	6574.727	6574.727
2027 Q2	6175.180	6175.180	6175.180
2027 Q3	5800.258	5800.258	5800.258
2027 Q4	5448.444	5448.444	5448.444

Forecast comparison table:

```
> future9.df
```

	Regression_Forecast	AR1_Residual_Forecast	Combined_Forecast
1	183176.1	9002.310	192178.5
2	170433.4	8453.146	178886.5
3	177006.3	7937.828	184944.2
4	176416.4	7454.271	183870.6
5	189744.7	7000.516	196745.2
6	177062.6	6574.727	183637.3
7	183696.2	6175.180	189871.3
8	183166.8	5800.258	188967.1
9	196555.8	5448.444	202004.3

#Question 3

The fixed ARIMA and auto ARIMA models were estimated on the training dataset and evaluated using the validation set. The auto ARIMA model produced lower RMSE, MAE, MAPE, and Theil's U compared to the fixed ARIMA model, indicating better forecasting accuracy. After selecting the best model, both models were re-estimated on the full dataset and used to produce nine-quarter forecasts. When comparing all models, the two-level regression + AR(1) residual model produced the lowest forecast errors overall, indicating it provided the best performance for this dataset.

Fixed Arima on training set. Forecast using validation partition:

```
> summary(arima.fixed)
Series: train.ts
ARIMA(1,1,1)(1,1,1)[4]

Coefficients:
      ar1      ma1      sar1      sma1
    0.1418  -0.2677  0.2199  -1.0000
s.e.  0.7079   0.6814  0.1729   0.1623

sigma^2 = 3278090:  log likelihood = -493.11
AIC=996.22  AICc=997.44  BIC=1006.26

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set -179.1361 1669.244 1126.504 -0.186929 0.9681938 0.2837019 -0.0151062

> arima.fixed.pred <- forecast(arima.fixed, h = nValid, level = 0)
> arima.fixed.pred
      Point Forecast      Lo 0      Hi 0
2021 Q1      140086.5 140086.5 140086.5
2021 Q2      144630.0 144630.0 144630.0
2021 Q3      142619.7 142619.7 142619.7
2021 Q4      157061.6 157061.6 157061.6
2022 Q1      144407.9 144407.9 144407.9
2022 Q2      149325.0 149325.0 149325.0
2022 Q3      147548.4 147548.4 147548.4
2022 Q4      161347.3 161347.3 161347.3
2023 Q1      148548.3 148548.3 148548.3
2023 Q2      153547.6 153547.6 153547.6
2023 Q3      151822.4 151822.4 151822.4
2023 Q4      165479.8 165479.8 165479.8
2024 Q1      152648.9 152648.9 152648.9
2024 Q2      157666.2 157666.2 157666.2
2024 Q3      155952.4 155952.4 155952.4
2024 Q4      169578.7 169578.7 169578.7
2025 Q1      156740.8 156740.8 156740.8
2025 Q2      161762.1 161762.1 161762.1
2025 Q3      160050.7 160050.7 160050.7
```


Auto Arima on training period, forecasting using validation partition:

```
> summary(arima.auto)
Series: train.ts
ARIMA(1,0,0)(1,1,0)[4] with drift

Coefficients:
      ar1      sar1      drift
    0.8233  -0.3287 1059.061
s.e.  0.0854   0.1539  273.276

sigma^2 = 4066426:  log likelihood = -504.68
AIC=1017.36  AICc=1018.15  BIC=1025.46

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set -51.98903 1895.26 1390.402 -0.05242751 1.211152 0.3501625 -0.02222372

> arima.auto.pred
      Point Forecast      Lo 0      Hi 0
2021 Q1      141448.6 141448.6 141448.6
2021 Q2      144831.1 144831.1 144831.1
2021 Q3      141324.4 141324.4 141324.4
2021 Q4      156917.8 156917.8 156917.8
2022 Q1      146999.8 146999.8 146999.8
2022 Q2      149913.3 149913.3 149913.3
2022 Q3      146246.9 146246.9 146246.9
2022 Q4      162165.1 162165.1 162165.1
2023 Q1      151799.3 151799.3 151799.3
2023 Q2      154691.1 154691.1 154691.1
2023 Q3      150932.4 150932.4 150932.4
2023 Q4      166624.6 166624.6 166624.6
2024 Q1      156307.9 156307.9 156307.9
2024 Q2      159126.0 159126.0 159126.0
2024 Q3      155331.1 155331.1 155331.1
2024 Q4      171042.8 171042.8 171042.8
2025 Q1      160664.9 160664.9 160664.9
2025 Q2      163470.1 163470.1 163470.1
2025 Q3      159656.5 159656.5 159656.5
```

Comparing accuracy of fixed arima vs auto arima:

```
> round(accuracy(arima.fixed.pred$mean, valid.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 6043.977 8998.679 7568.489 3.498 4.557 0.758    0.929

> round(accuracy(arima.auto.pred$mean, valid.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 5061.382 8444.12 6871.769 2.887 4.15 0.61    0.871
```

Arima fixed model on full data set:

```
> summary(arima.fixed.full)
```

Series: revenue.ts

ARIMA(1,1,1)(1,1,1)[4]

Coefficients:

	ar1	ma1	sar1	sma1
	0.4557	-0.5834	0.1373	-0.7321
s.e.	0.3834	0.3540	0.2637	0.2659

sigma^2 = 4623124: log likelihood = -671.98

AIC=1353.95 AICc=1354.84 BIC=1365.47

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	84.50456	2023.966	1424.491	0.0003819966	1.113934	0.2916879	0.0400178

Arima auto model on full data set:

```
> summary(arima.auto.full)
```

Series: revenue.ts

ARIMA(0,1,0)(0,1,1)[4]

Coefficients:

	sma1
	-0.6678
s.e.	0.1300

sigma^2 = 4552209: log likelihood = -672.93

AIC=1349.85 AICc=1350.02 BIC=1354.46

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	80.31847	2050.968	1480.204	0.008590517	1.15597	0.3030961	-0.06502539

Predictions 9 quarters ahead for both models using full dataset:

```
> arima.fixed.full.pred
      Point Forecast      Lo 0      Hi 0
2025 Q4      191149.4 191149.4 191149.4
2026 Q1      177684.9 177684.9 177684.9
2026 Q2      186649.9 186649.9 186649.9
2026 Q3      186967.1 186967.1 186967.1
2026 Q4      199078.3 199078.3 199078.3
2027 Q1      185982.7 185982.7 185982.7
2027 Q2      194634.7 194634.7 194634.7
2027 Q3      194742.3 194742.3 194742.3
2027 Q4      206932.0 206932.0 206932.0

> arima.auto.full.pred
      Point Forecast      Lo 0      Hi 0
2025 Q4      191743.7 191743.7 191743.7
2026 Q1      178809.6 178809.6 178809.6
2026 Q2      187777.0 187777.0 187777.0
2026 Q3      188047.2 188047.2 188047.2
2026 Q4      200294.9 200294.9 200294.9
2027 Q1      187360.9 187360.9 187360.9
2027 Q2      196328.3 196328.3 196328.3
2027 Q3      196598.4 196598.4 196598.4
2027 Q4      208846.2 208846.2 208846.2
```

Final comparing accuracy of 5 models: quadratic regression, two level model, fixed arima, auto arima, and seasonal naive - Two level model stands out as best.

```
> round((accuracy(reg.quad.full$fitted, revenue.ts)), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set  0 4733.615 4033.899 -0.15 3.328 0.863      0.484

> round((accuracy(reg.quad.full$fitted + fitted(ar1.full.res), revenue.ts)), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 183.389 1974.071 1502.192 0.113 1.203 0.013      0.188

> round((accuracy(arima.fixed.full$fitted, revenue.ts)), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 84.505 2023.966 1424.491  0 1.114 0.04      0.194

> round((accuracy(arima.auto.full$fitted, revenue.ts)), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 80.318 2050.968 1480.204 0.009 1.156 -0.065      0.197

> round((accuracy(snaive.full$fitted, revenue.ts)), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 4729.24 5882.997 4883.613 3.652 3.789 0.735      0.555
```